Exploring the Link Between Personality Traits and Substance Use

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Task 1. Setting up

Link to the repository: https://github.com/RichardMJaks/drug-use-analysis

We have given access to all instructors.

We have added this file to the repository in PDF format.

Task 2. Business understanding

Identifying your business goals

Background

Substance use among young and adult populations is a major public health concern. At the same time, personality traits and demographics are known to influence risk-taking, self-control, and social behaviour. Our project uses the "Drug Consumption Classification" dataset (1,885 respondents, Big Five scores, demographics, and self-reported use of 18 drugs) to examine how individual characteristics relate to drug use patterns. The project is rather exploratory.

Business goals

From a "business"/stakeholder perspective, our main goals are:

1. Better understand which personality traits and demographic factors are associated with different patterns of substance use.

2. Identify clusters of individuals with similar behavioural and consumption profiles that could, in principle, be targeted with different prevention or education strategies.

3. Produce clear visual and written outputs.

Business success criteria

The project is successful if:

- We can clearly describe at least a few robust relationships between personality traits/demographics and substance use categories, or clearly show no association between those two.

- We obtain interpretable clusters of individuals that can be meaningfully described (e.g. "low-risk occasional users", "polydrug sensation-seekers").
- The results and visualisations are understandable to non-technical people and can be summarised into a poster.

Assessing your situation

Inventory of resources

- **Data:** Kaggle "Drug Consumption Classification" dataset (Big Five scores, age, gender, education, country, ethnicity, 18 substances with recency labels).
- **People:** Course team members, instructors (for guidance).
- **Software/hardware:** Python (pandas, scikit-learn, visualisation libraries (ex Matplotlib)), Jupyter, GitHub repository for collaboration and version control.
- **Documentation:** CRISP-DM chapter used as process guideline.

Requirements, assumptions, and constraints

- Must follow CRISP-DM phases and deliver required reports within course deadlines.
- We assume that the dataset documentation on Kaggle correctly explains variable coding and that the sample is reasonably representative for the intended exploratory aims.
- Constraints: limited project time, no access to additional data, and limited domain expertise in clinical addiction research.
- The team must create a poster for the poster session in the University of Tartu Delta Centre, summarising the project's aims, methods, results, and conclusions in a clear and visually engaging format.
- All code, figures, and reports must be kept in the shared GitHub repository for transparency and version control.

Risks and contingencies

- **Risk:** Data quality issues (missing values, inconsistent encodings).

    - *Contingency:* Perform systematic data cleaning and, if necessary, simplify targets (e.g. binarise drug use outcomes).

- **Risk:** Models or clusters may be hard to interpret.

    - *Contingency:* Prioritise transparent methods (correlations, logistic regression, decision trees, simple clustering) and focus on qualitative interpretation, not just accuracy.

Terminology

We will maintain a short glossary for key terms, such as: "Big Five traits", "substance use recency categories", "cluster", "classifier", and "predictor variable", so that team members and readers share the same understanding, as recommended by CRISP-DM.

Costs and benefits

Direct financial costs are negligible (open data, free tools). Main costs are student time and computing resources. Benefits include learning CRISP-DM in practice, gaining insight into psychological predictors of drug use, and creating reusable code and visualisations.

Defining your data-mining goals

Data-mining goals

- Preprocess and clean the dataset (handle missing values, recode drug-use variables into meaningful classes).

- Quantify relationships between Big Five traits/demographics and use of specific drugs or drug categories (e.g. via correlation analysis and classification models).

- Cluster individuals based on personality and drug-use patterns to identify behavioural profiles.

- Produce visualisations (heatmaps, distribution plots, cluster plots) that support interpretation.

Data-mining success criteria

- Data cleaning and preparation scripts run reproducibly on the raw Kaggle data and produce an analysis-ready dataset.

- At least one classification model for selected drugs performs clearly better than a simple baseline (e.g. majority class) according to standard metrics (accuracy, F1 or AUC on a held-out set).

- Clustering yields a small number of stable clusters with distinct and interpretable profiles (checked by internal validation metrics and descriptive summaries).
- All steps are documented in the repository following the CRISP-DM deliverables for business understanding and data understanding.

Task 3. Data understanding

Gathering data

Outline data requirements

Our project requires:

- Personality trait scores (Big Five) to examine psychological predictors of drug use.
- Demographic variables such as age, gender, education, country, and ethnicity.
- Drug consumption variables for all 18 substances, ideally categorised by usage recency.
- Sufficient sample size to perform clustering, classification, and statistical analysis.
  The Kaggle dataset provides all required elements in a single CSV file.

Verify data availability

The Drug Consumption dataset is publicly available, downloadable without restrictions, and compatible with standard tools (Python/pandas, Jupyter). The dataset contains 1,885 rows and 32 attributes, including all Big Five traits, impulsivity, sensation seeking, demographic attributes, and drug-use recency indicators.
If we encounter unexpected technical or data-quality problems that prevent us from proceeding, we will consult with the instructors and, if necessary, switch to another dataset or project topic.

Define selection criteria

- Include all personality traits, all demographic variables, and all 18 drug-use variables, since they support all three project goals.
- Exclude no cases initially; outliers will be evaluated during exploration.

Describing data

- **Source:** Kaggle – Drug Consumption Classification dataset.
- **Format:** Single CSV file with 32 columns.

- **Cases:** 1,885 respondents.

- **Fields:**

  - **Personality traits:** NEO-FFI-R scores (Neuroticism, Extraversion, Openness, Agreeableness, Conscientiousness).
    - "Neuroticism is one of the Big Five higher-order personality traits in the study of psychology. Individuals who score high on neuroticism are more likely than average to be moody and to experience such feelings as anxiety, worry, fear, anger, frustration, envy, jealousy, guilt, depressed mood, and loneliness."
    - "Extraversion is one of the five personality traits of the Big Five personality theory. It indicates how outgoing and social a person is. A person who scores high in extraversion on a personality test is the life of the party. They enjoy being with people, participating in social gatherings, and are full of energy."x
    - "Openness is one of the five personality traits of the Big Five personality theory. It indicates how open-minded a person is. A person with a high level of openness to experience in a personality test enjoys trying new things. They are imaginative, curious, and open-minded. Individuals who are low in openness to experience would rather not try new things. They are close-minded, literal and enjoy having a routine."
    - "Agreeableness is one of the five personality traits of the Big Five personality theory. A person with a high level of agreeableness in a personality test is usually warm, friendly, and tactful. They generally have an optimistic view of human nature and get along well with others."
    - "Conscientiousness is one of the five personality traits of the Big Five personality theory. A person scoring high in conscientiousness usually has a high level of self-discipline. These individuals prefer to follow a plan, rather than act spontaneously. Their methodic planning and perseverance usually makes them highly successful in their chosen occupation." (https://www.kaggle.com/datasets/mexwell/drug-consumption-classification/data)

  - **Additional traits:** Impulsivity, Sensation seeking.
    - "In psychology, impulsivity (or impulsiveness) is a tendency to act on a whim, displaying behavior characterized by little or no forethought, reflection, or consideration of the consequences. If you describe someone as impulsive, you mean that they do things suddenly without thinking about them carefully first."
    - "Sensation is input about the physical world obtained by our sensory receptors, and perception is the process by which the brain selects, organizes, and interprets these sensations. In other words, senses are the physiological basis of perception."(https://www.kaggle.com/datasets/mexwell/drug-consumption-classification/data)

  - **Demographics:** Age category, gender, education level, country, ethnicity.

- **Drug consumption:** 18 substances coded with seven ordered categories (e.g., “Never used” to “Used in last day”).

The dataset is suitable for the project because it contains all the variables needed to test relationships between personality and drug use, build predictive models, and cluster individuals based on behavioural similarity. Its sample size is also adequate for clustering and classification tasks.

Exploring data

Initial observations

Values appear numeric at first, but they are all actually categorical, since every attribute has only a certain amount of different values. Distributions are roughly normal but vary slightly by trait. In Kaggle there is extra tables to translate these seemingly numeric values into categorical.

- Some drugs (e.g., crack, heroin) have strong class imbalance, which is important for classification modelling.

Initial findings / hypotheses

- Openness and sensation seeking appear higher among users of psychedelic drugs.
- Conscientiousness appears lower among users of stimulants and depressants.
- Clustering may reveal distinct groups such as “abstainers”, “experimenters”, and “polydrug users”.
  These are preliminary and require statistical testing.
- The subjects were picked unproportionally from the UK and USA, so this might not reflect generally over the western world.
- As expected, freely available and widely legal drugs have the most users, especially in daily or weekly categories.

Verifying data quality

CRISP-DM describes this step as confirming whether the data is adequate to support the project’s goals and identifying any severe issues, including missing or incorrect values, inconsistencies, or unusable features .

Quality checks

- **Missing values:**
  Initial inspection indicates very few or no missing values. Drug-use variables appear complete.

- **Inconsistent or incorrect values:**
  No invalid categorical values detected. Personality traits are within expected ranges.

- **Class imbalance:**
  Several substances show strong imbalance (e.g., crack, heroin, methadone). This must be addressed during modelling (e.g., through resampling or binarisation).
- Also research countries are imbalanced, subjects were mainly from UK and US

- **Encoding issues:**
  All variables use consistent formats, permitting easy cleaning and transformation.
- Types needed to be somewhat reformatted due to some issues with conversions

Conclusion on quality

The dataset contains **minor quality issues** but nothing severe enough to prevent analysis. The main challenges are class imbalance and categorical encodings, which will be addressed during data preparation. As CRISP-DM notes, quality issues should be documented and remedied before modelling to avoid unreliable results.

Task 4. Planning your project (0.25 points)

1) **Initial environment setup and workflow decisions**

Set up the analysis environment using Anaconda and Jupyter Notebook, agree on a consistent folder structure, and define a workflow that minimizes merge conflicts. Document initial data observations during setup.

Hours: Richard – 3h, Hannes – 3h.

2) **Data preparation and cleaning**

Load dataset, check formats, recode features into human-readable form (from continuous numbers into words and easily readable numbers).

Hours: Richard – 3h, Hannes – 3h.

3) **Data analysis**

Produce descriptive distributions, initial correlations, and demographic summaries. Summarize early findings for use in modelling and poster design by creating initial plots.

Hours: Richard – 7h, Hannes – 3h.

4) **Version control and documentation maintenance**

Keep GitHub repository organised (folders for notebooks, cleaned data, figures, reports). Write README updates and maintain clear commit messages.

Hours: Richard – 1h, Hannes – 1h.

5) **Clustering and pattern discovery (optional but highly aligned with your project goals)**

Apply clustering to identify behavioural subgroups. Summarise and visualise cluster characteristics.

Hours: Richard – 5h, Hannes – 1h.

6) **Version control and documentation maintenance**

Hours: Richard – 1h, Hannes – 1h.

7) **Visualization, report writing**

Create visualizations, write reports according to CRISP-DM guidelines.

Hours: Richard – 4h, Hannes – 8h.

8) **9. Final integration, quality control and submission**

Hours: Richard – 2h, Hannes – 2h.

9) **Create poster along with small script of bullet-points for poster presentation**

Design the poster for the Delta poster session

Hours: Richard – 1h, Hannes – 5h.

10) **Attendance at poster presentation**

Set up the poster. Talk with passersbys.

Hours: Richard – 3.5h, Hannes – 3.5h.

Extra (if something takes less time than anticipated): **Modelling**

Build predictive models to test whether a person's alcohol consumption status can predict their likelihood of using other drugs.